Practice Exam

MECH321 — Dynamics of Engineering Systems | TYUT 2026

Time allowed: 3 hours Total marks: 100

Answer all four questions. Question 1: ten multiple-choice items, 20 marks. Question 2: seven independent problems, 40 marks. Questions 3 and 4: 20 marks each.

Show essential steps and units. A scientific calculator is sufficient; MATLAB is not required. Supplied results may be used in later parts even if an earlier part is unfinished.

Question 1 Multiple choice 20 marks

Choose ONE best answer. Each item carries 2 marks; no partial credit.

1.1 A fixed-base mass is connected to a spring and a viscous damper. Which pair normally gives the two independent states? [2 marks]

A Applied force and damper force B Mass displacement and velocity

C Spring stiffness and damping coefficient D Mass and stored energy

1.2 Which statement describes a white-box model? [2 marks]

A It has no unknown parameters B It is fitted only from input and output measurements

C It is derived from physical laws with stated assumptions D It becomes grey-box whenever a prediction is verified

1.3 If the initial displacement is $x(0)=x_0$, which derivative transform is correct? [2 marks]

A $sX(s)$ B $X(s)/s$

C $sX(s)+x_0$ D $sX(s)-x_0$

1.4 A signal has transform $F(s)$. It starts 2 s later, with zero signal before then. What is its transform? [2 marks]

A $e^{-2s}F(s)$ B $e^{2s}F(s)$

C $F(s-2)$ D $F(s)/2$

Question 1 Multiple choice continued

1.5 A capacitor charges without an impulse current. Its voltage is initially 3 V. What does the initial value theorem give? [2 marks]

A The eventual voltage B The voltage time constant

C The voltage immediately after the switching instant is 3 V D The voltage must immediately fall to zero

1.6 For a zero-state system with transfer function $G(s)$, what is the output transform for a unit impulse input? [2 marks]

A $G(s)/s$ B $G(s)$

C $sG(s)$ D $1/G(s)$

1.7 A stable system has $G(s)=6/(s+3)$. What is its final output for a unit step? [2 marks]

A 2 B 3

C 6 D 0

1.8 Two given state equations are $\dot{x}_1=x_2$ and $\dot{x}_2=-5x_1-4x_2+u$. At $x_1=1$, $x_2=0$, $u=0$, which derivative pair is correct? [2 marks]

A $(0,2)$ B $(1,-3)$

C $(-2,0)$ D $(0,-5)$

1.9 Which pole pair gives a decaying oscillatory response? [2 marks]

A $1\pm j2$ B $-1\pm j2$

C $\pm j2$ D $0,2$

1.10 A stable unity-feedback system has one uncancelled integrator in its loop transfer function. What is its steady tracking error for a unit step? [2 marks]

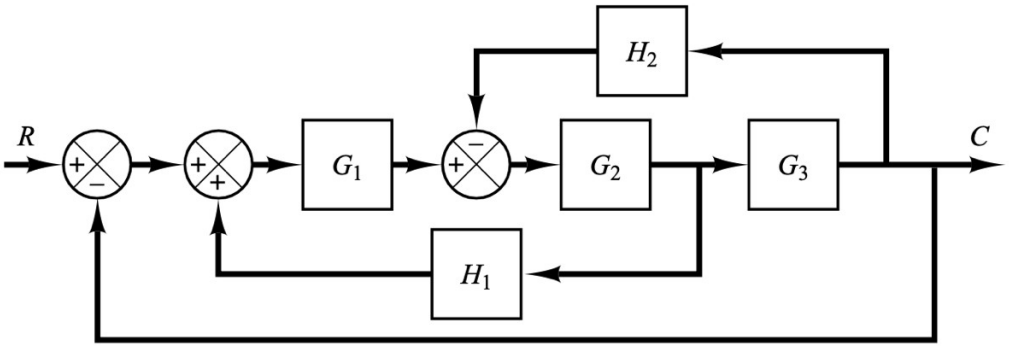
A It must be infinite B It equals the time constant

C Zero D It cannot be found without MATLAB

Question 2 Independent problems 40 marks

Each part is independent. Assume zero initial conditions when obtaining transfer functions.

(a) Find the overall transfer function $C(s)/R(s)$ for the block diagram below. Show the intermediate loop reductions or signal equations. Keep the blocks symbolic. [6 marks]



(b) A level deviation has transform $H(s) = 0.60/[s(s+3)]$, with level in metres. Use the final value theorem to find its eventual value. Briefly check that the theorem applies. [4 marks]

Question 2 continued

(c) A mechanical system obeys the following equation. Use the force–voltage analogy to write the corresponding electrical equation in charge q(t). Give numerical L, R and C with electrical units, and identify the counterparts of force, displacement and velocity. Do not draw a circuit. [4 marks]

$$2\ddot{x} + 6\dot{x} + 8x = f(t)$$

For the electrical model, use $i = \dot{q}$ and $L\ddot{q} + R\dot{q} + q/C = e(t)$.

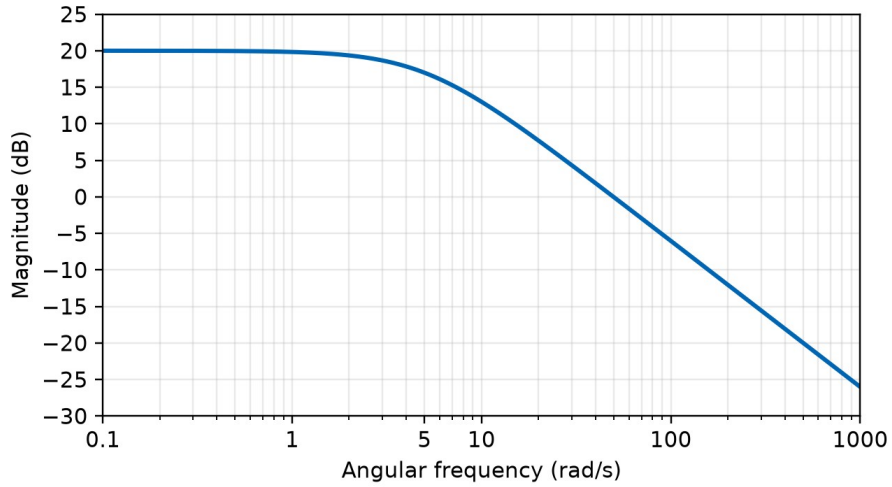
(d) A single tank has area A=0.50 m² and linear outlet resistance R=40 s/m². The variables h, q_i and q_o are deviations from an operating equilibrium: level (m), inflow and outflow (m³/s). Initially all deviations are zero. The inflow deviation steps to 0.010 m³/s at t=0. Use the supplied laws. [10 marks]

$$A\dot{h} = q_i - q_o, q_o = \frac{h}{R}$$

Write the numerical level ODE and the zero-state transfer function $H(s)/Q_i(s)$. Find the time constant. Use Laplace and inverse Laplace transforms to obtain h(t), showing the partial fractions. Find the time at which 90% of the final level CHANGE has occurred.

Question 2 continued

(e) The supplied magnitude plot is for a stable first-order system with positive DC gain and no zero. Read its low-frequency magnitude in dB and convert it to a gain factor. Estimate the corner frequency from the −3 dB point relative to the low-frequency level, and give the high-frequency asymptotic slope. Approximate graph readings are acceptable. [6 marks]



(f) A normalized system is given below. Time is in seconds; z and y have normalized output units, and u has the corresponding acceleration units. The coefficients 3 and 2 carry units s^{-1} and s^{-2}. Use the specified state order. [6 marks]

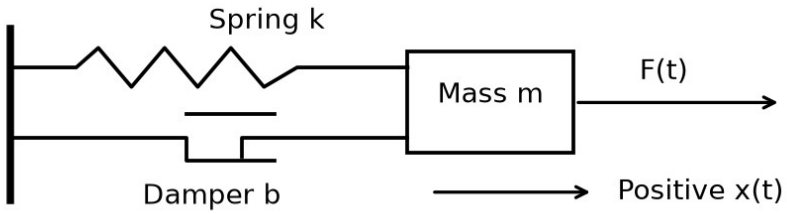
$\ddot{z} + 3\dot{z} + 2z = u$, $y = z$, $x_1 = z$, $x_2 = \dot{z}$

Write the two scalar state equations and the output equation. Give A, B, C and D in $\dot{x} = Ax + Bu$, $y = Cx + Du$, and state the model order.

(g) A loop has one open-loop right-half-plane pole. Its COMPLETE Nyquist trace encircles −1 once anticlockwise and does not pass through −1. There are no imaginary-axis open-loop poles. With clockwise encirclements positive, use $Z = P + N$ to find the number of closed-loop right-half-plane poles and decide stability. [4 marks]

Question 3 Mechanical step response 20 marks

A carriage starts at rest at its equilibrium position. Mass is 2 kg, spring stiffness is 8 N/m and viscous damping coefficient is 4 N s/m. A constant rightward force of 4 N is applied at time zero. Displacement from equilibrium is positive rightward. Assume a linear spring and damper and a fixed support.



(a) Write the numerical equation of motion using the stated positive direction. [5 marks]

(b) Find the natural frequency and damping ratio. State whether the system is underdamped, critically damped or overdamped. Use the supplied relationships below. [5 marks]

$$\omega_n = \sqrt{\frac{k}{m}}, \zeta = \frac{b}{2\sqrt{mk}}$$

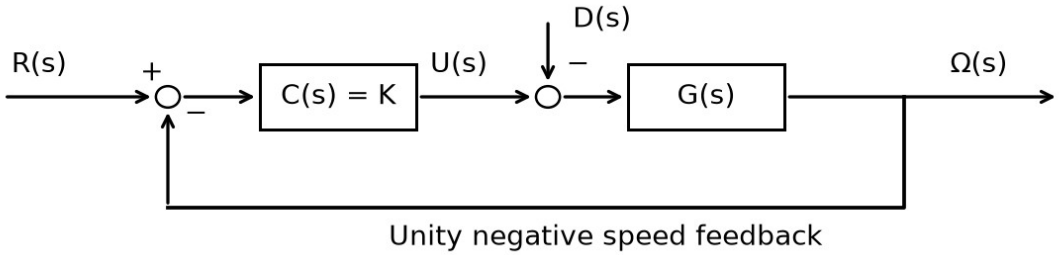
(c) Find the final displacement under the constant 4 N force. Explain briefly why the damping coefficient does not enter this final value. [5 marks]

(d) Use $x_\infty = 0.50$ m, $\zeta = 0.50$ and $\omega_n = 2$ rad/s. The fractional overshoot is supplied as $M_p = 0.163$. Calculate the peak displacement and approximate 2% settling time using the following formulas. [5 marks]

$$x_{peak} = x_\infty (1 + M_p), t_s \approx \frac{4}{\zeta \omega_n}$$

Question 4 Motor speed feedback 20 marks

A motor drives a flywheel with viscous rotational resistance $b = 0.50$ N m s/rad. The motor-torque-to-speed plant is supplied as $G(s) = \frac{2}{s+1}$. A proportional controller uses $u(t) = K[r(t) - \omega(t)]$ with gain 2 N m s/rad ($K = 2$ in SI values). The speed sensor is ideal, feedback is negative and unity, and initial speed is zero. The load torque $d(t)$ opposes rotation. Assume the actuator is unsaturated and there is no delay.



(a) Find the reference-to-speed closed-loop transfer function using the supplied plant and $K = 2$. [5 marks]

(b) Use the result $T(s) = \frac{4}{s+5}$. State the closed-loop pole and time constant, and whether the ideal loop is stable. [5 marks]

(c) For a constant reference of 3 rad/s and an opposing load of 0.50 N m, the steady balance is supplied below. Use $b = 0.50$ and $K = 2$ in SI values to find the final speed and the tracking error (reference minus actual speed). [5 marks]

$$(b + K)\omega_{\infty} = K r_0 - d_0$$

(d) At the gain crossover frequency, the loop phase is −75.52 degrees. Calculate and interpret the phase margin. [5 marks]

Use these relationships where helpful:

$$T(s) = \frac{KG(s)}{1 + KG(s)}, PM = 180^{\circ} + \angle L(j\omega_{gc})$$

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